- Egrets usually are present near cattle. They catch insects which are disturbed by the cattle. The bird benefits, while the cattles are not affected (**Fig. 11.13).**



Fig. 11.13 Egrets near cattle; barnacles on whale

COMPETITION (–, –): It refers to the type of interaction in which individuals of a species or members of different species vie for limited availability of food, water, nesting space, cover, mates or other resources. When resources are in more than adequate to meet the demands of the organisms seeking them, competition does not occur, but when inadequate to satisfy the need of the organisms seeking them, the weakest, least adapted, or least aggressive individuals are often forced to face challenges. This phenomenon is known as the **competitive exclusion** principle of Hardin.

DEGREE OF COMPETITION - Competition is usually keen between individuals of the same species (intraspecific competition) because they have identical requirements for food, mates, and so on. Interspecific competition occurs where different species require at least some resource materials or conditions in common. The severity of competition depends on the extent of similarity or overlap in the requirements of different individuals and the shortage of the supply in the habitat as birds compete with squirrels for nuts, and seeds; insects and ungulates compete for food in grasslands (**Fig. 11.14).**

PARASITISM (+, –): It is a kind of harmful interaction between two species, wherein one species is the 'parasite' and the other its 'host'. The parasite benefits at the expense of the host. A parasite derives shelter, food and protection from the host. Parasites exhibit adaptations to exploit their hosts. The parasites may be viral parasites (plant / animal viruses), microbial parasites (e.g., bacteria / protozoa / fungi), phytoparasites (plant parasites) and zooparasites (animal parasites such as Platyhelminthes, nematodes, arthropods). Parasites may inhabit or attach to the surface of the host (Ectoparasites - Head lice, Leech) or live within the body of the host (endoparasites – ascaris, tapeworm). The endoparasites usually live in the alimentary tract, body cavities, various organs or blood or other tissues of the host.



Fig. 11.14 Owls competing for food

Parasites may be permanent or temporary. Temporary parasites spend only a part of their life cycle as parasites. For example, Glochidium larva of *Anadonia* (fresh water mussel) attaches itself to the body of fish. Permanent parasites spend their life completely dependent on their host organism. The common examples of permanent parasites are *Plasmodium*, *Entamoeba*, Round worms, Pin worms, Tape worms, etc.,

PREDATION (+, –): It is a form of interaction, where one animal kills another animal for food. Like parasitism, predation is important in community dynamics, but both differ in the point that a predator tends to be larger than its prey, and it catches its prey from without, while a parasite is smaller than its host and consumes it from within.

XII Std Zoology Chapter 11 EM.indd 192 23/12/2022 16:51:13